

SYSTEM REQUIREMENTS SPECIFICATION

Pre-Owned Car Fair Market Valuation Engine (Random Forest ML)

Document Ref: SRS-ML-2026-V1
Target Domain: Automotive Retail
Model Type: Ensemble Regression

Project Scope & Problem Statement

An automotive dealership network requires a machine-learning solution to predict the fair market resale value of pre-owned cars based on core vehicle attributes (make, model year, mileage, fuel type, transmission, engine capacity, and accident history). The project analyzes historical sales data, cleans and encodes features, develops a baseline Decision Tree Regressor alongside an ensemble Random Forest Regressor, compares performance metrics (RMSE, MAE, R²), and extracts feature importances.

1. BUSINESS REQUIREMENTS (BRD)

BR-1: Automated Valuation Engine

Build a machine-learning pipeline to calculate fair market resale values based on vehicle specifications, eliminating subjective manual estimation.

BR-2: Ensemble Gain Quantification

Benchmark model performance to quantify the exact accuracy improvement provided by an ensemble Random Forest Regressor over a single Decision Tree Regressor.

BR-3: Feature Drivers Analysis

Identify and rank key vehicle attributes (e.g., mileage vs. accident history) to explain price variations transparently to buyers and sellers.

BR-4: Operational Model Delivery

Deliver a finalized valuation model that can reliably predict prices on new pre-owned car inventory with minimal prediction error.

2. RECOMMENDED KAGGLE DATASETS

To match the required features (make, year, mileage, fuel, transmission, engine capacity, accident history), the following Kaggle datasets are recommended:

DATASET NAME	FEATURES AVAILABLE	ALIGNMENT WITH PROBLEM STATEMENT
USA Cars Dataset (Kaggle: doaaalsenani/usa-cers-dataset)	Price, Brand (Make), Year, Mileage, Fuel, Transmission, Title Status (Accident/Clean)	Primary Choice: Explicitly includes title status (salvage/clean) to represent accident history alongside key vehicle specs.
Used Car Price Prediction Dataset (Kaggle: pushpakhinglaspure/used-car-price)	Make, Model, Year, Mileage, Fuel_Type, Transmission, Engine (CC), Power, Price	Secondary Choice: Features engine capacity in CC, making it ideal for engine displacement modeling.

3. DATA & MODEL PIPELINE

1. DATA CLEANING Impute nulls & strip units	2. ENCODING One-Hot & Ordinal encoding	3. BASELINE MODEL Decision Tree Regressor	4. ENSEMBLE MODEL Random Forest Regressor	5. METRICS & IMPORTANCES RMSE, MAE, R ² & Feature
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4. FUNCTIONAL REQUIREMENTS (FR)

REQ ID	MODULE	DETAILED FUNCTIONAL DESCRIPTION
FR-01	Feature Ingestion & Cleaning	The system shall ingest the 7 target features (make, model year, mileage, fuel type, transmission, engine capacity, accident history), handle missing values, and calculate <code>vehicle_age</code> ($\text{Current Year} - \text{Model Year}$).
FR-02	Feature Encoding	The system shall apply One-Hot Encoding or Ordinal Encoding to categorical variables (fuel type, transmission, make) and scale numerical inputs where necessary.
FR-03	Baseline Modeling	The system shall train a single Decision Tree Regressor on the preprocessed training set to establish performance benchmark metrics.
FR-04	Ensemble Modeling	The system shall train an ensemble Random Forest Regressor with multi-tree bagging, tuning hyperparameters (e.g., <code>n_estimators</code> , <code>max_depth</code> , <code>min_samples_split</code>).
FR-05	Model Comparison	The system shall compute and compare performance across both models on an unseen test set using three evaluation metrics: RMSE, MAE, and R^2 .
FR-06	Feature Importance Analysis	The system shall extract and rank Gini / MDI feature importances from the Random Forest model to identify top drivers of vehicle resale value.

5. NON-FUNCTIONAL REQUIREMENTS (NFR)

NFR-1: Reproducibility

All data splits and random model initializations must use a fixed seed (`random_state=42`) to ensure 100% reproducible results.

NFR-2: Performance Improvement

The ensemble Random Forest model must demonstrate a lower RMSE and higher R^2 score compared to the baseline Decision Tree.

NFR-3: Inference Speed

The valuation model must return price predictions within 100 milliseconds for a single vehicle attribute input vector.

NFR-4: Model Interpretability

Feature importances must be clearly exportable/visualizable so analysts can verify that key features (e.g., mileage, age) logically influence price.

6. COMPARATIVE EVALUATION METRICS

The evaluation module will compare the Decision Tree Regressor against the Random Forest Regressor using the following metrics:

METRIC	FORMULA	DESCRIPTION & TARGET
RMSE (Root Mean Squared Error)	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	Measures standard deviation of residuals. Lower is better (Random Forest should minimize large errors).
MAE (Mean Absolute Error)	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	Represents the average error magnitude in price units. Lower is better.
R^2 Score (Coefficient of Determination)	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Measures proportion of variance explained by vehicle features. Target: $R^2 \geq 0.85$.